

TalentMatch AI: Enterprise Workforce Intelligence

An AI-Driven Semantic Matching and Autonomous Team Assembly
System

Bayan Marashdeh

Abstract

Traditional recruitment and workforce allocation systems rely heavily on exact keyword matching, leading to inefficiencies and suboptimal team formations. This project introduces TalentMatch.ai, an enterprise-grade AI agent designed to revolutionize workforce intelligence. By leveraging Vector Search (ChromaDB) for semantic understanding and Machine Learning (Logistic Regression) for predictive success scoring, the system evaluates true candidate capabilities. Furthermore, it introduces autonomous team assembly, skill gap analysis, and a natural language AI agent to orchestrate complex HR operations.

Contents

1	Introduction	4
1.1	Problem Statement	4
1.2	Proposed Solution	4
2	System Architecture	4
3	Core System Features	4
3.1	Talent Scout (Semantic Matcher)	4
3.2	Autonomous Team Assembly	5
3.3	Skill Gap Analysis	6
3.4	Talent AI Agent	6
4	Mathematical Foundation	7
5	Conclusion	7

1 Introduction

1.1 Problem Statement

Current human resource management systems suffer from a rigid dependency on exact keyword searches. When a project manager requires a "Data Scientist," traditional databases might overlook a highly qualified candidate whose resume lists "Machine Learning Engineer" or specific Python libraries, simply due to a lack of semantic understanding.

1.2 Proposed Solution

TalentMatch AI solves this by representing candidate skills and job requirements as dense vectors. This allows the system to compute semantic similarities rather than relying on overlapping text. The solution integrates a FastAPI backend, a Streamlit frontend, and a suite of AI tools to automate candidate scoring, team building, and training recommendations.

2 System Architecture

The system follows a modern, decoupled architecture designed for scalability and rapid prototyping:

- **Frontend (Streamlit):** Selected for its ability to rapidly deploy interactive, data-heavy dashboards without the overhead of building a React/Vue interface from scratch.
- **Backend API (FastAPI):** Chosen for its high performance, asynchronous capabilities, and automatic interactive API documentation (Swagger UI).
- **Vector Database (ChromaDB):** Implemented to store and query text embeddings. It is lightweight and highly optimized for semantic search operations compared to traditional relational databases.
- **Relational Database (PostgreSQL):** Used to maintain structured, relational records such as employee IDs, historical projects, and training catalogs.

3 Core System Features

3.1 Talent Scout (Semantic Matcher)

The core feature of the system is the semantic matching engine. It replaces standard SQL LIKE queries with vector similarity search.

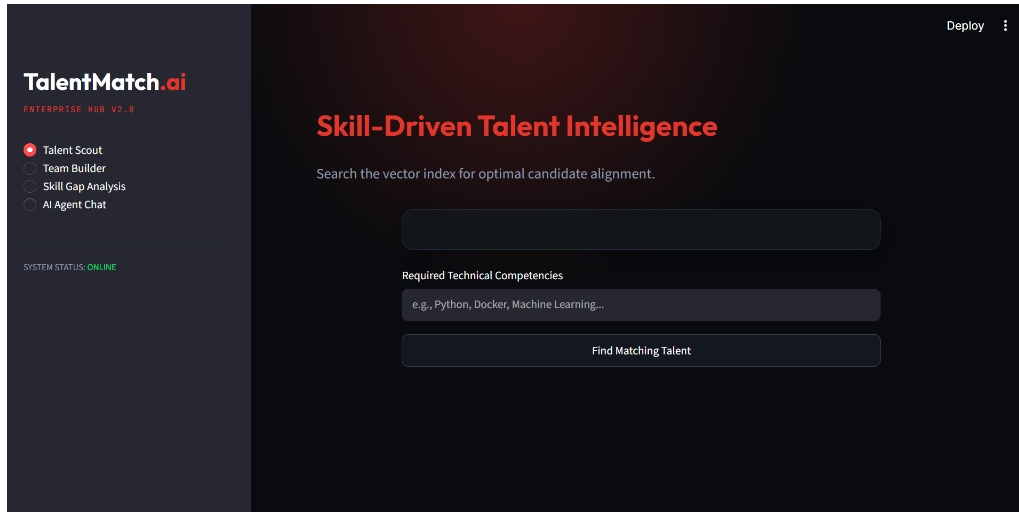


Figure 1: Talent Scout Interface displaying match results.

Design Choice: A Logistic Regression model was trained to predict candidate success. This model was chosen over complex deep learning models (like Neural Networks) because of its *explainability*. In HR applications, knowing *why* a candidate received an 88.5% score (feature weights like skill match and experience) is critical for compliance and trust.

3.2 Autonomous Team Assembly

This endpoint receives a project brief and a required team size, then algorithmically selects a combination of employees.

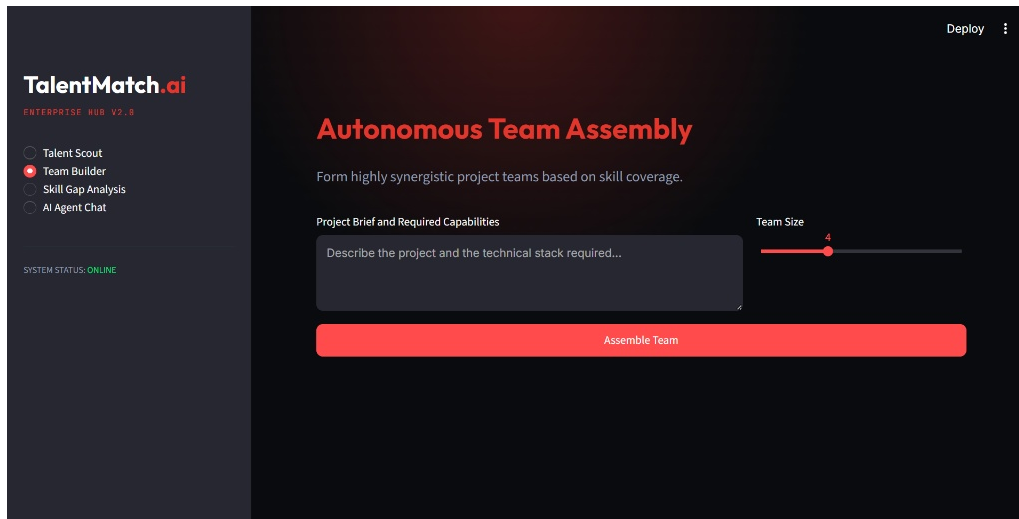


Figure 2: Autonomous Team Assembly Interface.

Design Choice: Instead of finding the single best employee four times (which might result in redundant skills), the algorithm calculates the *Combined Skill Coverage*. It ensures that the collective skills of the selected individuals cover the entirety of the project requirements, mimicking human managerial decisions.

3.3 Skill Gap Analysis

To support internal mobility, this module compares an employee's current proficiencies against the requirements of a target role.

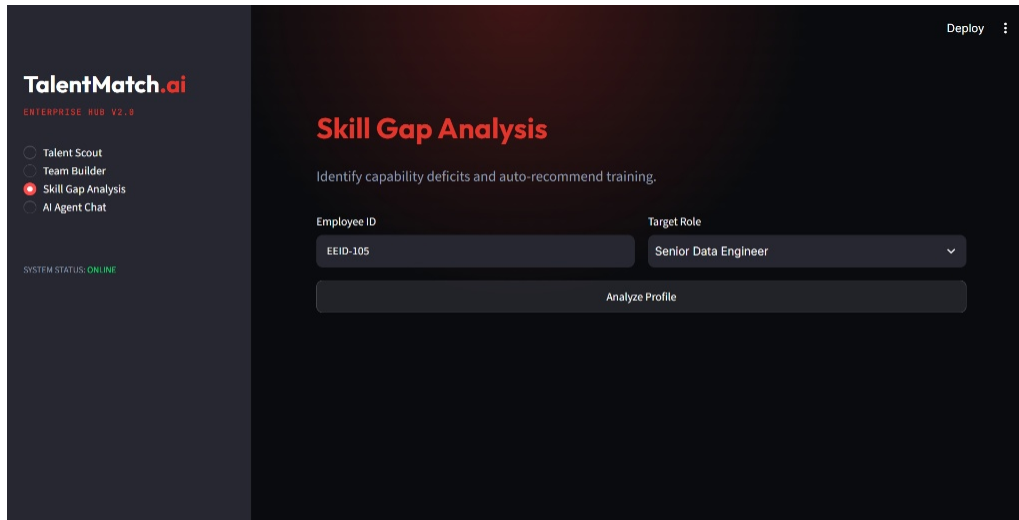


Figure 3: Skill Gap Analysis and Training Recommendations.

Design Choice: The system integrates a simulated Retrieval-Augmented Generation (RAG) approach. Once a skill deficit is identified (e.g., missing "AWS"), it queries the training catalog database to recommend the exact course needed, moving the system from a passive analytical tool to an actionable decision-support system.

3.4 Talent AI Agent

A natural language interface designed to orchestrate the backend tools based on user prompts.

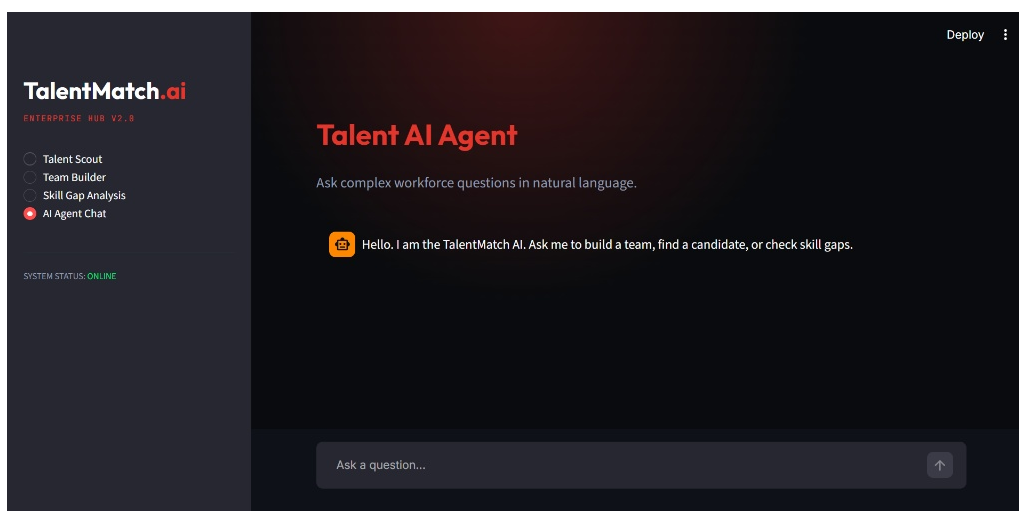


Figure 4: Conversational AI Agent Interface.

Design Choice: We implemented an Intent Routing mechanism. If a user asks to "build a team," the agent triggers the `build_team()` function. Crucially, guardrails were

implemented: if queried about unrelated topics (e.g., "weather"), the agent refuses to answer, demonstrating strict adherence to its domain as an Enterprise HR Tool.

4 Mathematical Foundation

The success probability of a candidate is calculated using the Logistic Function:

$$P(Y = 1|X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \dots + \beta_n X_n)}} \quad (1)$$

Where X represents normalized features such as skill overlap percentage, years of experience, and project relevance.

5 Conclusion

TalentMatch.ai successfully demonstrates the viability of vector-based semantic search and machine learning in modernizing workforce intelligence. By providing explainable matching scores, autonomous team assembly, and actionable upskilling recommendations, the system offers a comprehensive alternative to keyword-based HR platforms.